

ORIGINAL ARTICLE

Association between quantitative HBsAg and serum HBV DNA in occult HBV infection

SRY WARDA OKTAVYA¹, MANSYUR ARIF¹, ASVIN NURULITA^{1,2}, LIONG BOY KURNIAWAN¹, LUTHFI PAREWANGI^{2,3}, ULENG BAHRUN¹

¹Department of Clinical Pathology, Faculty of Medicine, Hasanuddin University, Makassar, Indonesia; ²Wahidin Sudirohusodo Hospital, Makassar, Indonesia; ³Department of Internal Medicine, Faculty of Medicine, Hasanuddin University, Makassar, Indonesia.

ABSTRACT

Background and aim: Negative hepatitis B surface antigen (HBsAg) testing does not necessarily exclude hepatitis B virus (HBV) persistence, particularly in endemic regions where occult hepatitis B infection (OBI) may remain undetected. Because HBV DNA testing may be limited by cost and laboratory capacity, quantitative HBsAg (qHBsAg) has been proposed as a potential surrogate marker. However, its diagnostic utility in OBI remains uncertain due to suppressed antigen expression. This study aimed to evaluate the association between qHBsAg and serum HBV DNA among adults with OBI.

Methods: An observational cross-sectional study was conducted at a tertiary referral hospital in Makassar, Indonesia. Adults (≥ 18 years) with non-reactive HBsAg and detectable serum HBV DNA were included. HBV DNA was quantified using real-time polymerase chain reaction, and qHBsAg was measured using an automated chemiluminescent immunoassay.

Results: Fifty-five participants were included (mean age 54.69 ± 11.79 years; 67.3% male), including 30.9% with hepatic malignancy. qHBsAg values were uniformly non-reactive and narrowly distributed (0.00074–0.01958 IU/mL), whereas HBV DNA remained detectable across the cohort (1.0000–1.9400 IU/mL). qHBsAg showed no significant correlation with HBV DNA ($r = -0.160$; $p = 0.242$), and the association remained non-significant after adjustment ($r = -0.098$; $p = 0.490$).



Received: 16 February 2026 | Accepted: 16 March 2026

Correspondence: Sry Warda Oktavya, MD / Department of Clinical Pathology, Faculty of Medicine, Hasanuddin University, Makassar, 90245, Indonesia / Jl. Perintis Kemerdekaan KM 10, 90245, Makassar, Indonesia
E-mail: srywardao@gmail.com

Conclusion: In individuals with OBI, qHBsAg within the non-reactive range does not reflect inter-individual variability in circulating HBV DNA and therefore has limited utility as a surrogate marker of viraemia. These findings support the continued need for HBV DNA testing for accurate molecular confirmation of occult HBV infection. (www.actabiomedica.it)

Key words: occult hepatitis B infection, quantitative HBsAg, hepatitis B virus DNA, HBsAg-negative, Indonesia.

Introduction

Negative HBsAg testing is widely interpreted as evidence against ongoing hepatitis B virus (HBV) infection in routine clinical care (1). Yet in a clinically relevant subset of individuals, HBV persistence can remain detectable at the molecular level despite absent surface antigen in serum. This phenomenon, termed occult hepatitis B infection (OBI), challenges conventional HBV screening paradigms and complicates risk assessment in populations with previous HBV exposure (2). Clinically, the HBsAg-negative serologic profile is particularly relevant because it frequently represents past infection, but may also mask persistent infection with potential implications for liver disease progression, hepatocellular carcinoma (HCC), and HBV reactivation under immunosuppression (3, 4). HBV DNA testing remains the most direct approach to confirm OBI. Nevertheless, this test is not uniformly accessible in resource-limited settings and may be constrained by cost, laboratory capacity, and availability of standardized assays (5). In this context, there is a practical clinical need for alternative biomarkers that could help clinicians identify or triage individuals who require confirmatory molecular testing (6). Several serological and molecular markers have therefore been explored to complement HBV DNA testing in the evaluation of occult infection. These include anti-HBc serology and emerging molecular markers reflecting intrahepatic viral activity such as HBV RNA or cccDNA-related transcripts. However, many of these markers remain limited by availability, technical complexity, or uncertain clinical interpretation in individuals with suppressed antigen expression (7, 8).

Quantitative HBsAg (qHBsAg) has been extensively evaluated in overt chronic HBV infection, where it is often discussed as a marker related to antigen production and viral transcriptional activity (9, 10). Several studies have investigated alternative biomarkers and serological profiles in individuals with OBI, highlighting the persistence of viral DNA despite absent or extremely low levels of circulating HbsAg (11-13). These features raise an important question for clinical care in endemic settings: can qHBsAg be used as a simpler, lower-cost surrogate marker to estimate viral persistence or circulating viraemia in individuals with suspected OBI. However, the biological and analytical context of OBI differs fundamentally from overt chronic HBV infection. In Indonesia, where HBV exposure remains common and confirmatory molecular testing is not routinely accessible in all settings, the interpretation of serologically atypical profiles has practical clinical implications (14, 15). By definition, surface antigen expression in OBI is suppressed to extremely low levels that may approach the lower analytical limit of detection of commercial assays (16). Under these conditions, qHBsAg measurements may cluster near the assay's lower limit of quantification, thereby reducing discriminatory capacity (11). If this occurs, qHBsAg would be unable to reflect inter-individual differences in viraemia even when HBV DNA remains detectable. Importantly, evidence on the relationship between qHBsAg and HBV DNA in HBsAg-non-reactive individuals remains limited, particularly in Indonesia, where tertiary referral cohorts often include patients with advanced liver disease and hepatic malignancy, making diagnostic interpretation and risk stratification clinically relevant (17, 18). Therefore,

this study aimed to evaluate the association between qHBsAg and serum HBV DNA among adults fulfilling an operational definition of OBI at a tertiary referral hospital in Makassar, Indonesia. In addition, we explored whether the observed relationship differed by sex and diagnostic category (hepatic malignancy versus non-malignant liver disease). Clarifying the clinical utility or limitation of qHBsAg in this context is important to inform diagnostic strategies and determine whether qHBsAg can contribute to risk stratification or triage of HBV DNA testing in endemic settings.

Methods

Study design, setting, and participants

This observational analytical cross-sectional study evaluated the association between quantitative HBsAg (qHBsAg) and serum HBV DNA using stored serum specimens obtained from Wahidin Sudirohusodo Hospital, Makassar, Indonesia. Laboratory analyses were conducted at the Hasanuddin University Medical Research Center (HUM-RC) and the UDD PMI South Sulawesi laboratory. Demographic and clinical information, including age, sex, and clinical diagnostic category, were obtained from medical records. Diagnostic status was classified as hepatic malignancy or non-malignant liver disease based on the clinician's documented diagnosis. Stored serum specimens were eligible for inclusion if obtained from adults aged ≥ 18 years with non-reactive HBsAg based on routine serological testing and detectable serum HBV DNA. Samples were excluded if the available specimen volume was insufficient for laboratory analyses. The minimum sample size was estimated using the Lemeshow

formula for prevalence estimation as a conservative approach, assuming an event proportion of 2% ($P = 0.02$), 95% confidence level ($Z = 1.96$), and a margin of error of 5% ($d = 0.05$), yielding a minimum sample size of approximately 30 participants. A total of 55 eligible specimens were included in the final analysis. The study was approved by the Health Research Ethics Committee of the Faculty of Medicine, Hasanuddin University–Wahidin Sudirohusodo Hospital, Makassar (Ethical Clearance No. 2/UN4.6.4.5.31/PP36/2026), under the parent ethical approval (No. 348A/UN4.6.4.5.31/PP36/2025).

HBV DNA measurement

Serum HBV DNA was quantified by real-time polymerase chain reaction (real-time PCR). Viral DNA was extracted from serum using the Zymo Research DNA extraction kit (Zymo Research, USA) in accordance with the manufacturer's protocol. The extracted DNA was subsequently used as the template for HBV DNA amplification and quantification. qHBsAg was measured using a fully automated chemiluminescent enzyme immunoassay (CLIA) platform (Sysmex HISCL Automated Immunoassay System) with the HISCL HBsAg Assay Kit (Sysmex Corporation, Japan), based on a two-step sandwich immunoassay principle. qHBsAg results were reported in IU/mL, and values < 0.03 IU/mL were interpreted as non-reactive. Only samples with detectable HBV DNA were included in the analysis according to the assay output.

Statistical analysis

Descriptive statistics were used to summarize participant characteristics and laboratory measurements.

Table 1. Participant characteristics.

Variables			
Age (years old) mean $\pm$ SD		54.69	11.79
Sex (n / %)	Male	37	67.3
	Female	18	32.7
Clinical diagnosis (n / %)	Hepatic malignancy	17	30.9
	Non-malignant hepatic disease	38	69.1

Table 2. Distribution of qHBsAg and HBV DNA values in the current study.

Variables	Minimum	Maximum	Median	Mean	SD
qHBsAg (IU/mL)	0.00074	0.01958	0.00089	0.0017	0.0033
HBV DNA (IU/mL)	1.0000	1.9400	1.2100	1.2349	0.6242

Continuous variables are presented as mean $\pm$ standard deviation and median (minimum–maximum), while categorical variables are presented as frequency (n) and percentage. The association between qHBsAg and HBV DNA was assessed using Spearman correlation. To account for potential confounding, partial correlation analysis was also performed controlling for age, sex, and diagnostic status. Statistical analyses were performed using a standard statistical software package. A two-sided p-value <0.05 was considered statistically significant.

Results

Participant characteristics and baseline laboratory findings

Fifty-five adults fulfilling the study's operational definition of OBI were included. The mean age was 54.69 ± 11.79 years (median 55; range 32–85), indicating that the study population largely represented middle-aged to older adults. The cohort was predominantly male (37/55, 67.3%), with females accounting for 18/55 (32.7%). With respect to clinical diagnosis, 38/55 (69.1%) were categorized as non-malignant liver disease, whereas 17/55 (30.9%) had hepatic malignancy (Table 1). qHBsAg concentrations were consistently within a very narrow and non-reactive range (0.00074–0.01958 IU/mL; mean 0.0017 ± 0.0033 IU/mL; median 0.00089 IU/mL). In contrast, HBV DNA was detectable across the cohort (range 1.0000–1.9400 IU/mL; mean 1.2349 ± 1.6242 IU/mL; median 1.2100 IU/mL) (Table 2). These findings align with profiles reported in previous studies of low-level HBV replication despite non-reactive HBsAg levels.

Association between qHBsAg and HBV DNA

The relationship between qHBsAg and serum HBV DNA was assessed using Spearman correlation

in 55 participants. qHBsAg demonstrated a weak inverse association with HBV DNA in bivariate analysis ($r_b = -0.160$; $p = 0.242$), which remained weak and non-significant after adjustment for age, sex, and diagnostic category in partial correlation analysis ($r = -0.098$; $p = 0.490$) (Table 3). Consistent with the correlation analysis, scatterplot visualization showed a dispersed distribution without a discernible monotonic trend (Figure 1). The fitted linear trend line is shown for visualization only and explained minimal variance ($R^2 = 0.013$). Data points largely clustered at very low qHBsAg levels despite measurable variability in HBV DNA, and the presence of a small number of higher qHBsAg values did not establish a consistent association. Stratified scatterplots similarly showed no apparent association across clinically relevant subgroups. When stratified by sex (Figures 2), both male and female participants demonstrated broadly dispersed data points with clustering of qHBsAg values near the lower assay range, while HBV DNA values varied across the observed spectrum. The fitted trend lines in both strata were nearly flat and did not suggest a consistent relationship between qHBsAg and HBV DNA. Likewise, stratification by diagnostic category (Figures 3) showed comparable patterns in participants with hepatic malignancy and those with non-malignant liver disease. In both diagnostic subgroups, qHBsAg remained concentrated at very low levels and did not exhibit a discernible monotonic or linear trend with HBV DNA. Taken together, these visual patterns were consistent with the correlation analyses and indicate that the absence of association between qHBsAg and HBV DNA was robust across sex- and diagnosis-defined subgroups. Descriptively, mean qHBsAg was 0.0012 IU/mL in the hepatic malignancy group and 0.0019 IU/mL in the non-malignant group, while mean HBV DNA values were comparable between groups (both approximately 1.23 IU/mL) (Table 4).

Discussion

In this study of adults with HBsAg-non-reactive serology and detectable HBV DNA fulfilling an operational definition of occult hepatitis B infection (OBI), quantitative HBsAg values were uniformly non-reactive and clustered within a very narrow range, whereas HBV DNA remained detectable across all participants. Correlation analyses demonstrated no significant association between qHBsAg and serum HBV DNA levels, and this finding remained unchanged after adjustment for age, sex, and diagnostic category. Stratified analyses similarly showed no

meaningful relationship between the two parameters across clinically relevant subgroups. These findings suggest that within the biological context of OBI, qHBsAg measured within the non-reactive range has limited capacity to reflect inter-individual variability in circulating viraemia. From a clinical perspective, these findings underscore the limitations of HBsAg-based screening when used in isolation (16, 19). Despite uniformly non-reactive quantitative HBsAg, molecular evidence of viral persistence remained detectable, indicating that HBsAg negativity does not necessarily reflect virological clearance, particularly in endemic settings (20, 21). HBV DNA values clustered within a constrained range across participants, suggesting limited inter-individual variability in circulating viraemia within this cohort. These laboratory findings were consistent with the study’s operational definition of OBI, defined as detectable serum HBV DNA in the presence of non-reactive HBsAg (22). Notably, nearly one-third of participants carried a diagnosis of hepatic malignancy, highlighting the potential clinical relevance of occult infection in populations at risk for

Table 3. Correlation between qHBsAg and serum HBV DNA.

	HBV DNA			
	r_b	p	r_p	p
qHBsAg	-0.160	0.242	-0.098	0.490

r_b = bivariate correlation coefficient; r_p = partial correlation coefficient (controlling for age, gender, and diagnosis status); p = p-value.

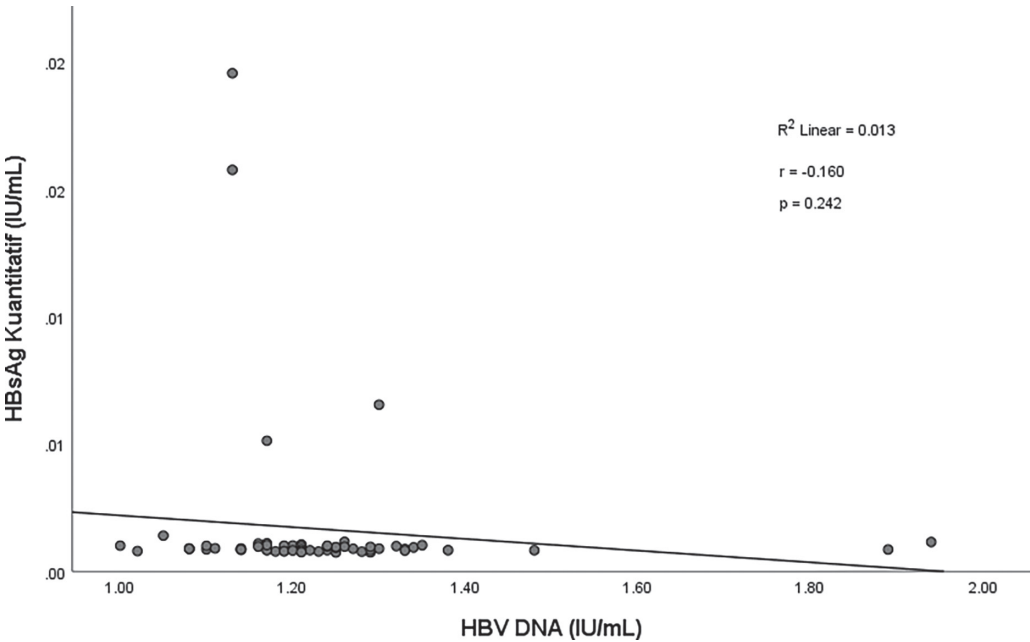


Figure 1. Relationship between qHBsAg and serum HBV DNA levels in the study cohort. Scatterplot showing qHBsAg (IU/mL) plotted against serum HBV DNA (IU/mL) among 55 participants with HBsAg non-reactive serology. The solid line represents the fitted linear regression trend. The association was weak and non-significant (Spearman correlation $r = -0.160$; $p = 0.242$), with minimal explained variance ($R^2 = 0.013$).

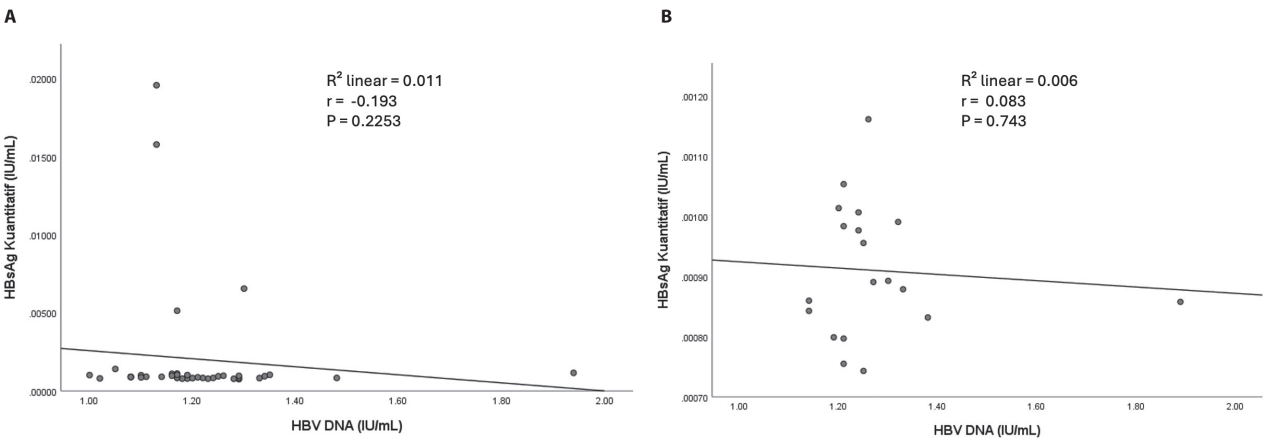


Figure 2. Association between qHBsAg and serum HBV DNA stratified by sex. Scatterplots showing qHBsAg (IU/mL) plotted against serum HBV DNA (IU/mL) in (A) male and (B) female participants. The solid line represents the fitted linear regression trend. Both subgroups demonstrated minimal explained variance ($R^2 = 0.011$ in males; $R^2 = 0.006$ in females), indicating no meaningful linear association between qHBsAg and HBV DNA by sex.

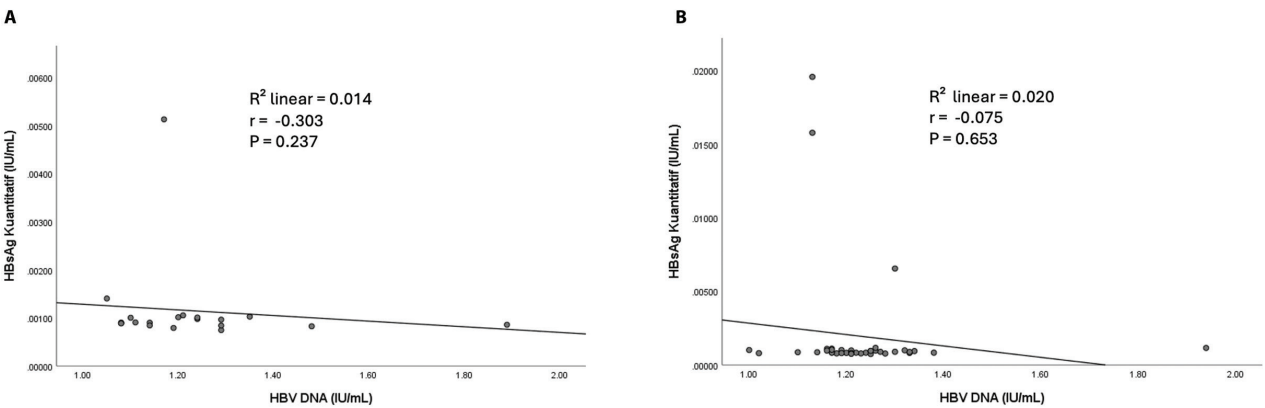


Figure 3. Association between qHBsAg and serum HBV DNA stratified by clinical diagnosis. Scatterplots showing qHBsAg (IU/mL) plotted against serum HBV DNA (IU/mL) in (A) hepatic malignancy and (B) non-malignant liver disease groups. The solid line represents the fitted linear regression trend. Explained variance remained negligible in both subgroups ($R^2 = 0.014$ and $R^2 = 0.020$, respectively), supporting a consistently weak relationship between qHBsAg and HBV DNA across diagnostic categories.

Table 4. qHBsAg and serum HBV DNA levels by diagnostic category

	qHBsAg (IU/mL)		HBV DNA (IU/mL)	
	mean±SD	range	mean±SD	range
Hepatic malignancy	0.0012±0.00	0.004	1.23±0.20	0.84
Non-malignant hepatic disease	0.0019±0.00	0.019	1.23±0.14	0.94

advanced liver outcomes. While causality cannot be inferred from this cross-sectional analysis, the coexistence of detectable HBV DNA with HBsAg negativity provides biological plausibility for the hypothesis that occult viral persistence may remain relevant in hepatocarcinogenesis and liver disease progression (22, 23). It is also possible that underlying malignancy or associated immune dysregulation could influence viral markers or antigen expression. However, in the present study, stratified analyses demonstrated similarly weak and non-significant relationships between qHBsAg and HBV DNA across diagnostic categories, suggesting that the presence of hepatic malignancy did not materially alter the observed association. Together, these baseline data support the integration of HBV DNA testing into clinical evaluation pathways for non-reactive HBsAg individuals, especially those with suspected chronic liver disease or malignancy, where viral classification may influence monitoring strategies and risk stratification. In this cohort of HBsAg-non-reactive individuals with detectable HBV DNA, qHBsAg did not correlate with circulating HBV DNA levels, even after adjustment for age, sex, and diagnostic category. This finding is clinically important because qHBsAg is often discussed as a potential marker reflecting transcriptional activity of intrahepatic cccDNA and antigen production in overt chronic HBV infection (8). However, our data suggest that in the context of OBI, qHBsAg values clustered at the assay's lower range and therefore offered little discriminatory capacity to reflect inter-individual differences in viraemia. The extremely low explained variance in scatterplots further supports that any linear relationship between the two parameters, if present, is biologically weak and unlikely to be clinically meaningful. The observed decoupling of qHBsAg and HBV DNA is consistent with the biological definition of OBI, in which viral persistence may continue in the absence of measurable surface antigen in peripheral blood (7). In such settings, circulating HBsAg becomes an unreliable proxy of viral activity, likely reflecting suppressed antigen expression and/or secretion rather than absence of replicative templates (24). Importantly, the absence of correlation was reproducible across sex and diagnostic strata, indicating that the lack of association was not driven by demographic imbalance or by the

presence of hepatic malignancy. This strengthens the generalizability of the observation within this clinical population and supports the view that the relationship between HBsAg expression and viraemia fundamentally differs between overt HBV infection and occult states. Several limitations should be considered when interpreting these findings. First, this was a cross-sectional study using stored serum specimens, which limits causal inference and prevents evaluation of temporal or longitudinal changes in HBV DNA and qHBsAg levels. Second, the study cohort was derived from a single tertiary referral hospital and included clinically heterogeneous patients, including those with hepatic malignancy, which may limit generalizability to community or screening populations. Third, most qHBsAg values clustered close to the assay's lower quantification range, which reduces discriminatory capacity and may partly explain the lack of correlation with HBV DNA. Finally, the sample size was relatively small ($n = 55$) and HBV DNA values were distributed within a narrow range, which may have limited statistical power to detect weak associations. Larger multicentre studies with standardized serological profiles, longitudinal follow-up, and additional virological markers are warranted to refine the clinical utility of qHBsAg in serologically atypical HBV infection. Future studies should incorporate larger sample sizes, longitudinal follow-up, and standardized assays with clearly defined lower limits of detection and quantification to better characterize viral dynamics in OBI. Inclusion of intrahepatic markers (e.g., cccDNA, HBV integration signatures) and broader immunologic or virologic profiling may further clarify mechanisms underlying the dissociation between surface antigen expression and viraemia. Such data will be critical to refine diagnostic strategies, improve risk stratification, and guide clinical decision-making in patients with atypical HBV serology. In conclusion, in this cross-sectional study of HBsAg-non-reactive adults with detectable HBV DNA, qHBsAg values clustered within a narrow non-reactive range while HBV DNA remained detectable across participants. qHBsAg showed no significant correlation with serum HBV DNA levels in bivariate analysis, and this remained unchanged after adjustment for age, sex, and diagnostic category. Stratified analyses by sex and hepatic malignancy status similarly

demonstrated no meaningful association. Collectively, these findings indicate that in occult hepatitis B infection, qHBsAg does not reflect inter-individual variability in circulating HBV DNA and therefore has limited utility as a surrogate marker of viraemia.

Ethic Approval: The study was approved by the Health Research Ethics Committee of the Faculty of Medicine, Hasanuddin University–Wahidin Sudirohusodo Hospital, Makassar (Ethical Clearance No. 2/UN4.6.4.5.31/PP36/2026) under the parent approval (No. 348A/UN4.6.4.5.31/PP36/2025).

Conflict of Interest: Each author declares that he or she has no commercial associations (e.g., consultancies, stock ownership, equity interest, patent/licensing arrangement, etc.) that might pose a conflict of interest in connection with the submitted article.

Authors' Contribution: SWO: Laboratory analyses, Writing – original draft; MA, AN: Study design, Data interpretation, Writing – review and editing; LBK, LP: Clinical methodology, Supervision; U: Data validation, Study coordination. All authors reviewed and approved the final version of the manuscript and agreed to be accountable for all aspects of the work.

Declaration on the Use of AI: Artificial intelligence tools were used only for language editing and grammar improvement.

References

1. Guvenir M, Arikan A. Hepatitis B virus: from diagnosis to treatment. *Pol J Microbiol.* 2020;69(4):391-9. doi:10.33073/pjm-2020-044.
2. Soriano V, González R, Barea L, et al. Occult hepatitis B infection: unveiling clinical burden and new diagnostic strategies. *Expert Rev Gastroenterol Hepatol.* 2026;1-10. doi:10.1080/17474124.2025.2611857.
3. Raimondo G, Allain JP, Brunetto MR, et al. Statements from the Taormina expert meeting on occult hepatitis B virus infection. *J Hepatol.* 2008;49(4):652-7. doi:10.1016/j.jhep.2008.07.014.
4. Mak L-Y, Wong DK-H, Pollicino T, Raimondo G, Hollinger FB, Yuen M-F. Occult hepatitis B infection and hepatocellular carcinoma: epidemiology, virology, hepatocarcinogenesis and clinical significance. *Journal of Hepatology.* 2020;73(4):952-64. doi:10.1016/j.jhep.2020.05.042.
5. Wang C, Xue R, Wang X, Xiao L, Xian J. High-sensitivity HBV DNA test for the diagnosis of occult HBV infection: commonly used but not reliable. *Front Cell Infect Microbiol.* 2023;13:1186877. doi:10.3389/fcimb.2023.1186877.
6. Yang Y, Xing H, Wu G, Zhao Y. Global prevalence of occult hepatitis B virus infection in HBcAb-positive individuals: a systematic review and meta-analysis. *Scientific Reports.* 2025;15(1):44244. doi:10.1038/s41598-025-23207-4.
7. Saitta C, Pollicino T, Raimondo G. Occult hepatitis B virus infection: an update. *Viruses.* 2022;14(7). doi:10.3390/v14071504.
8. Testoni B, Scholtès C, Plissonnier ML, et al. Quantification of circulating HBV RNA expressed from intrahepatic cccDNA in untreated and NUC-treated patients with chronic hepatitis B. *Gut.* 2024;73(4):659-67. doi:10.1136/gutjnl-2023-330644.
9. Islam S, Shahera U, Jahan M, Tabassum S. Evaluation and determination of quantitative hepatitis B surface antigen diagnostic performance in chronic hepatitis B virus-infected patients. *Cureus.* 2023;15(6):e41202. doi:10.7759/cureus.41202.
10. Yılmaz T, Özdemir E, Düzençi D, Bahçecioglu İH. Clinical and histopathological correlation of quantitative HBsAg levels in chronic hepatitis B. *International Journal of Hepatology.* 2025;2025(1):9096871. doi:10.1155/ijh/9096871.
11. Kuhns MC, Holzmayer V, Anderson M, et al. Molecular and serological characterization of Hepatitis B virus (HBV)-positive samples with very low or undetectable levels of HBV surface antigen. *Viruses.* 2021;13(10):2053.
12. Bubonja-Šonje M, Peruć D, Abram M, Mohar-Vitezić B. Prevalence of occult hepatitis B virus infection and characterisation of hepatitis B surface antigen mutants among adults in western Croatia. *Ann Hepatol.* 2024;29(1):101156. doi:10.1016/j.aohep.2023.101156.
13. Zhong G, Jiang Z-H, Wang X-Y, et al. Increasing prevalence of occult HBV infection in adults vaccinated against hepatitis B at birth. *Vaccines.* 2025;13(2):174.
14. Lusida MI, Juniastuti, Yano Y. Current hepatitis B virus infection situation in Indonesia and its genetic diversity. *World J Gastroenterol.* 2016;22(32):7264-74. doi:10.3748/wjg.v22.i32.7264.
15. Yamani LN, Triani E, Amin M, et al. Prevalence and genotype distribution of hepatitis B virus among migrant workers in Lombok Island, Indonesia. *Asian Pacific Journal of Tropical Medicine.* 2020;13(1).
16. Jaroszewicz J, Serrano BC, Wursthorn K, et al. Hepatitis B surface antigen (HBsAg) levels in the natural history of hepatitis B virus (HBV) infection: a European perspective. *Journal of Hepatology.* 2010;52(4):514-22. doi:10.1016/j.jhep.2010.01.014.
17. Nwokediuko SC, Osuala PC, Uduma UV, Alaneme AK, Onwuka CC, Mesigo C. Pattern of liver disease admissions in a Nigerian tertiary hospital. *Niger J Clin Pract.* 2013;16(3):339-42. doi:10.4103/1119-3077.113458.
18. Loho IM, Hasan I, Lesmana CRA, Dewiasty E, Gani RA. Hepatocellular carcinoma in a tertiary referral hospital in Indonesia: lack of improvement of one-year survival rates between 1998-1999 and 2013-2014. *Asian Pacific Journal of Cancer Prevention.* 2016;17(4):2165-70.

19. Wang H, Wang J, Zhang S, et al. Clinical features and transition of acute hepatitis B virus infection. *Journal of Viral Hepatitis*. 2025;32(2):e14048. doi:10.1111/jvh.14048.
20. Ali H, Buenning C, Daghfal D. Advancing HBV diagnostics: the role of ultrasensitive HBsAg testing. *Diagnostics (Basel)*. 2025;15(21). doi:10.3390/diagnostics15212744.
21. Posuwan N, Wasitthankasem R, Pimsing N, et al. Hepatitis B prevalence in an endemic area of hepatitis C virus: a population-based study implicated in hepatitis elimination in Thailand. *Journal of Virus Eradication*. 2024;10(4):100577. doi:10.1016/j.jve.2024.100577.
22. Delghandi S, Raoufinia R, Shahtahmasbi S, Meshkat Z, Gouklani H, Gholoobi A. An overview of occult hepatitis B infection (OBI) with emphasis on HBV vaccination. *Heliyon*. 2024;10(17):e37097. doi:10.1016/j.heliyon.2024.e37097.
23. Nishikawa H, Osaki Y. Clinical significance of occult hepatitis B infection in progression of liver disease and carcinogenesis. *J Cancer*. 2013;4(6):473–80. doi:10.7150/jca.6609.
24. Mirian M, Taghizadeh R, Khanahmad H, et al. Exposition of hepatitis B surface antigen (HBsAg) on the surface of HEK293T cell and evaluation of its expression. *Res Pharm Sci*. 2016;11(5):366–73. doi:10.4103/1735-5362.192485.